

INTERNATIONAL CYBERSECURITY AND  
DIGITAL FORENSICS ACADEMY

## Week 1 Practical Laboratory

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### Information Security Governance in Action

*GlobalHealth Connect (GHC) — Executive Governance Simulation*

|                        |                                                          |
|------------------------|----------------------------------------------------------|
| Course                 | GRC102 – Information Security Governance                 |
| Module                 | Module 1 – Principles of Information Security Governance |
| Access Window          | 05–11 September 2026                                     |
| Due                    | 11 September 2026, 11:59 PM WAT                          |
| Maximum Score          | 100 marks                                                |
| Gradebook Contribution | 10% – Practical Laboratories                             |

**Submission mode: Written report + supporting evidence**

*Scenario-based lab • No terminal or command-line work is required for this laboratory*

## 1. Laboratory Overview

You have been appointed Director of Information Security Governance at GlobalHealth Connect (GHC), a mid-sized health-technology organisation that provides cloud-based patient management systems. Following rapid growth, two acquisitions and a contained data-leakage near-miss, the Board has determined that GHC's current ad-hoc security approach is no longer sustainable.

This practical laboratory places you inside a realistic governance environment. Rather than performing technical command-line tasks, you will analyse organisational information, respond to executive requests, design governance artefacts, interpret security metrics and build a practical roadmap that turns a reactive security programme into a business-aligned governance function.

## 2. Learning Objectives

- Design an information security governance structure appropriate to a growing organisation.
- Define accountability using organisational structures and a RACI responsibility model.
- Draft a formal Information Security Charter that establishes mandate, authority and scope.
- Translate technical security information into concise Board-level metrics and risk reporting.
- Design a Security Steering Committee that supports cross-functional decision-making.
- Assess governance maturity and propose a measurable 12–18 month improvement roadmap.

## 3. Key Stakeholders

| Stakeholder     | Role                     | Primary Concerns                                               |
|-----------------|--------------------------|----------------------------------------------------------------|
| Sarah Chen      | Chief Executive Officer  | Growth, reputation, operational efficiency and accountability. |
| Marcus Thorne   | Chief Financial Officer  | Return on investment, compliance cost and enterprise risk.     |
| Elena Rodriguez | Chief Technology Officer | Developer productivity, technical debt and innovation.         |
| David Miller    | Board Member             | Accountability, meaningful metrics and regulatory compliance.  |

## 4. Rules of Engagement

- Treat all scenario information as confidential organisational information for the purpose of this simulation.
- Base decisions on governance principles, business context and evidence provided in this brief.
- Where external sources are used, cite them appropriately. Do not fabricate laws, standards, evidence or metrics.
- AI tools may be used for brainstorming or language support, but you remain responsible for verification, reasoning and originality. Any material AI assistance should be declared in your submission.
- Submit only work you can explain and defend during grading or moderation.

## 5. Task 1 Governance Blueprint

**Executive request:** The CEO requires a formal security governance structure with clear roles, reporting relationships and accountability following the recent near-miss.

### Current Organisational Context

| Employee / Role                   | Department         | Relevant Context                                              |
|-----------------------------------|--------------------|---------------------------------------------------------------|
| John Smith – IT Manager           | IT Operations      | Manages infrastructure and performs some security activities. |
| Jane Doe – Senior Developer       | Development        | Leads a key product development team.                         |
| Mark Johnson – Compliance Officer | Legal & Compliance | Focuses on healthcare regulatory obligations.                 |
| Emily White – Data Analyst        | Finance            | Handles sensitive financial information.                      |
| Robert Green – HR Manager         | Human Resources    | Owns onboarding and offboarding processes.                    |

### Required Evidence Task 1

1. A current-state governance gap assessment identifying the most important weaknesses in GHC's ad-hoc model.
2. A proposed security governance organisational chart showing the Board, CEO, governance function and relevant business/technology relationships.
3. A RACI matrix covering at least six governance activities, including policy approval, risk review and incident response planning.
4. A short explanation of how the proposed structure supports accountability, transparency, business alignment and risk management.

## 6. Task 2 Information Security Charter

The CFO and Board require a formal mandate for the information security programme. Draft a concise but complete Information Security Charter that can be approved by executive management and referenced across the organisation.

### GHC Strategic Goals for 2026

| Strategic Goal                 | Business Intent                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------|
| Expand Market Share            | Increase the client base by 25% through new features and entry into two regional markets. |
| Enhance Operational Efficiency | Reduce overhead by 10% and improve system uptime to 99.99%.                               |
| Maintain Customer Trust        | Protect patient information and aim for zero major data breaches.                         |
| Foster Innovation              | Invest in AI-driven diagnostics and secure data-exchange capabilities.                    |
| Achieve Regulatory Excellence  | Meet applicable healthcare and data-protection obligations proactively.                   |

### Charter Sections

- Purpose — why the Information Security Programme exists and how it supports GHC.
- Scope — assets, systems, data, people and third parties covered by the programme.

- Authority — source of authority, decision rights and escalation powers.
- Roles and Responsibilities — high-level governance accountabilities linked to Task 1.
- Key Principles — risk-based, business-aligned, proportionate and continuously improving security.
- Reporting Structure — management, committee and Board reporting expectations.
- Review and Approval — approval authority, review frequency and change control.

### Required Evidence Task 2

5. Completed GHC Information Security Charter.
6. A 1–2 paragraph justification memo to Marcus Thorne, CFO, explaining how the Charter supports GHC's strategic goals, accountability and investment decisions.

## 7. Task 3 Board Reporting and Security Metrics

The Board has requested a concise, business-focused security update. Select the most decision-useful metrics from the six-month dataset, interpret the trends and present the organisation's security posture without unnecessary technical jargon.

| Month     | Phishing | Malware | Critical Vulns Patched | Patching % | Training % | High-Risk Incidents |
|-----------|----------|---------|------------------------|------------|------------|---------------------|
| September | 150      | 25      | 10 of 15               | 67%        | 45%        | 2                   |
| October   | 180      | 30      | 12 of 20               | 60%        | 50%        | 3                   |
| November  | 210      | 35      | 15 of 25               | 60%        | 55%        | 4                   |
| December  | 250      | 40      | 18 of 30               | 60%        | 60%        | 5                   |
| January   | 280      | 45      | 20 of 35               | 57%        | 65%        | 6                   |
| February  | 320      | 50      | 22 of 40               | 55%        | 70%        | 7                   |

### Required Evidence Task 3

7. A one-page Board Executive Summary containing an overall Red/Amber/Green posture assessment and justification.
8. Five selected security metrics showing current status, six-month trend and concise business commentary.
9. At least two priority risks/issues and two actionable recommendations.
10. A short rationale explaining why your five selected metrics are suitable for Board oversight.

## 8. Task 4 Security Steering Committee

A dispute has emerged between the CTO and IT Manager regarding a proposed password policy. The CTO believes the policy will create excessive friction; the IT Manager argues that stronger controls are necessary. The CEO wants a formal Security Steering Committee (SSC) to resolve such cross-functional issues.

**Conflict context:** The proposed policy requires 16-character complex passwords changed every 30 days without password-manager integration. The CTO recommends a more balanced approach emphasising multi-factor authentication and secure password-management.

### Required Evidence Task 4

11. Security Steering Committee Terms of Reference covering purpose, scope, membership, responsibilities, meeting cadence, decision authority and reporting.
12. A sample agenda for the first SSC meeting with the password-policy conflict as a key discussion item.
13. A 1–2 paragraph briefing note to the CEO explaining how the SSC will resolve cross-functional conflict and maintain strategic alignment.

## 9. Task 5 Governance Maturity Assessment

The Board wants evidence that the new governance arrangements will improve over time. Assess GHC's current maturity using the simplified 1–5 model below and propose a realistic roadmap to achieve at least Level 3 (Defined) across all domains within 12–18 months.

### Current-State Interview Summary

| Governance Domain        | Current-State Evidence                                                                  |
|--------------------------|-----------------------------------------------------------------------------------------|
| Policy & Documentation   | Policies are inconsistent, inherited from acquisitions and not always current.          |
| Roles & Responsibilities | Security ownership and accountability are unclear.                                      |
| Risk Management          | The organisation is reactive and lacks a formal proactive assessment process.           |
| Metrics & Reporting      | Technical metrics exist, but they are not consistently translated into business impact. |
| Training & Awareness     | Annual training is mandatory, but engagement is low.                                    |
| Compliance               | Preparation is audit-driven rather than continuously monitored.                         |

### Simplified Maturity Scale

| Level          | Description                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------|
| 1 – Ad Hoc     | Informal, reactive and inconsistent; no defined processes.                                    |
| 2 – Initial    | Some processes exist but are inconsistently documented or applied.                            |
| 3 – Defined    | Processes are documented, standardised, communicated and supported by clear responsibilities. |
| 4 – Managed    | Processes are measured, monitored and reviewed against performance targets.                   |
| 5 – Optimising | Continuous improvement is embedded and processes adapt quickly to changing risk.              |

### Required Evidence Task 5

14. A completed maturity assessment for Policy & Documentation, Roles & Responsibilities, Risk Management, Metrics & Reporting, Training & Awareness, and Compliance.
15. A 12–18 month maturity roadmap containing 3–5 priority initiatives. For each initiative, state the objective, key activities and measurable expected outcome.
16. A 1–2 paragraph executive summary to the Board explaining GHC's current maturity and the strategic importance of the roadmap.

## 10. Submission Package

Submit one professionally organised report containing the five task evidence bundles. Supporting diagrams, charts, matrices and other evidence may be embedded in the report or attached as supporting evidence where the E-Campus workflow permits.

- Task 1: governance gap assessment, organisational chart, RACI matrix and governance rationale.
- Task 2: Information Security Charter and CFO justification memo.
- Task 3: Board Executive Summary, five metrics and metric-selection rationale.

- Task 4: SSC Terms of Reference, sample meeting agenda and CEO briefing note.
- Task 5: maturity assessment, 12–18 month roadmap and Board executive summary.

### Submission Standards

- Preferred report format: PDF or DOCX.
- Use clear headings and label each task and evidence item.
- Charts, matrices and diagrams must be legible and professionally presented.
- Cite external sources using a consistent academic referencing style.
- Recommended filename: GRC102\_W1\_Lab\_[StudentName]\_[RegistrationNumber].pdf
- Due: 11 September 2026 at 11:59 PM WAT.

## 11. Assessment Criteria

| Criterion           | Weight | What Assessors Will Look For                                                            |
|---------------------|--------|-----------------------------------------------------------------------------------------|
| Completeness        | 20%    | All five task evidence bundles are complete and address the required outputs.           |
| Clarity & Coherence | 25%    | Clear structure, logical reasoning and executive-quality communication.                 |
| Strategic Alignment | 20%    | Recommendations reflect GHC's business goals, stakeholders and risk context.            |
| Critical Thinking   | 20%    | Decisions are justified, practical and supported by appropriate governance reasoning.   |
| Professionalism     | 15%    | High-quality presentation, terminology, grammar, referencing and evidence organisation. |

**Maximum Score: 100 marks | Gradebook Contribution: 10% | Practical Pass: institutional standard**

## 12. Academic Integrity and Evidence Authenticity

All submitted evidence must represent the student's own analysis and decision-making. Fabricated organisational evidence, invented citations, copied work, or submission of AI-generated material that the student cannot explain may be treated as an academic integrity concern. Students must preserve enough working evidence to support their submission if requested during moderation.

## Appendix A — GHC Company Context

GlobalHealth Connect is a provider of cloud-based patient-management systems serving healthcare organisations. Its business depends on customer trust, protection of sensitive health information, regulatory compliance, innovation and continued growth through product development and acquisition.

| Business Driver | Governance Relevance                                                                                                 |
|-----------------|----------------------------------------------------------------------------------------------------------------------|
| Trust           | Security governance must protect patient information and organisational reputation.                                  |
| Compliance      | Governance should ensure responsibilities, evidence and oversight for applicable healthcare and privacy obligations. |
| Innovation      | Security decisions should enable safe development of AI-enabled and digital-health capabilities.                     |
| Growth          | Governance must remain effective as the organisation expands and integrates acquisitions.                            |

## Appendix B — Governance Terms

| Term               | Working Definition                                                                                 |
|--------------------|----------------------------------------------------------------------------------------------------|
| Governance         | The system by which an organisation is directed and controlled.                                    |
| Policy             | A high-level statement of organisational intent and direction.                                     |
| Standard           | A mandatory requirement or specification supporting policy.                                        |
| Procedure          | A defined sequence of steps for performing an activity.                                            |
| RACI               | A responsibility model identifying Responsible, Accountable, Consulted and Informed roles.         |
| Steering Committee | A cross-functional body that provides strategic direction, prioritisation and oversight.           |
| Maturity Model     | A structured method for assessing the consistency and sophistication of processes or capabilities. |